



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Spring 2023

Paper: Computational Physics - I

Course Code: PHYS-3502

Roll No.

Time: 3 Hrs.

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions:

(6x5=30)

- I. Write a C++ program to print the given series and sum of the series: $S = \sum_{n=1}^{10} \frac{(-1)^n x^{n/2}}{n(n+1)}$, ask user to input value of x.
- II. Write C++ program with function to calculate $f(x) = 5x^2 - 2x - 1$ and print values for x and f(x). Implement for five iterations.
- III. Write a C++ program to calculate and print factorial of a number taken from user using while loop.
- IV. Write a C++ program to generate random numbers in a range of 0 to N.
- V. Write a C++ program to find a trace of matrix of order 4x4.
- VI. Write a C++ program to find a transpose of matrix of order 3x3.

Q.2. Answer the following questions.

(3x10=30)

- I. Write a C++ program to solve the differential equation $\frac{dy}{dx} = x^3 - y$ subject to the initial condition $y(0)=1$ and $h=0.05$ on the interval $[0,0.2]$.
- II. Write a C++ program to use the Trapezoidal rule with $n=6$ to estimate the integral $I = \int_1^2 \frac{1}{x} dx$.
- III. Write a C++ program to evaluate the $\frac{dy}{dx} = x + y; y(0) = 1, [0,0.5], h = 0.1$ by Runge-Kutta method (4th order).

PUACP

Fifth Semester - Spring

2023:

Short Question:

(i)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
clrscr();
```

```
int n, sum=0; x;
```

```
cout << "Enter value of x=";
```

```
cin >> x;
```

```
for (n=1; n<10; n++)
```

```
{
```

```
sum += pow(-1, n) * pow(x, n/2) /  
(n * (n+1));
```

```
}
```

```
cout << "Final result of series="
```

```
<< sum;
```

```
getch();
```

```
}
```


(ii)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
void main()
```

```
{
```

```
float x; float a[5];
```

```
for (int i=0; i<5; i++)
```

```
{
```

```
cout << "Enter the value of x=";
```

```
cin >> a[i];
```

```
x = 5*a[i]*a[i] - 2*a[i] - 1;
```

```
cout << "f(x) = |t|" << x << endl;
```

```
}
```

```
getch();
```

```
}
```

(iii)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```

    {
        int n;
        int fact = 1;
        int i = 1;
        cout << "Enter the number: << endl;
        cin >> n;
        while (i <= n)
        {
            fact = fact * i;
            i = i + 1;
        }
        cout << "The factorial of
        number = " << fact << endl;
        getch();
    }

```

(iv)

```

#include <iostream.h>
#include <conio.h>
#include <math.h>
void main()
{

```



```

for (int i = 0; i <= N; i++)
{
    cout << "Random numbers = ";
    face = 1 + rand() % 6;
    if (i % 5 == 0)
        cout << endl;
}
getch();
}

```

(v) x

```

#include <iostream.h>
#include <conio.h>
void main()
{
    int a[4][4], i, j;
    cout << "Enter the elements of matrix : " << endl;
    for (i = 0; i < 4; i++)
        for (j = 0; j < 4; j++)
        {
            cin >> a[i][j];
        }
}

```

```

    }
    cout << "You have entered a matrix" << endl;
    for (i=0; i<4; i++)
    {
        for (j=0; j<4; j++)
        {
            cout << a[i][j] << endl;
        }
        cout << endl;
    }
    getch();
}

```

(vi)

```

#include <iostream.h>
#include <conio.h>
void main ()
{
    int A[3][3];
    cout << "Enter A = ";
    for (int i=0; i<3; i++)

```



```

for (j=0; j<3; j++)
    cin >> A[i][j];
cout << "\n A= \n";
for (int i=0; i<3; i++)
{
    for (int j=0; j<3; j++)
    {
        cout << A[i][j];
    }
    cout << endl;
}

```

// transpose of A.

```

for (int i=0; i<3; i++)
{
    for (int j=0; j<3; j++)
        A[i][j] = A[j][i];
    cout << endl;
}
getch();

```

Other Method:

S-Q (V) → 2023

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{
```

```
int a[4][4];
```

```
int i, j, trace = 0;
```

```
cout << "Enter elements of  
the matrix:"
```

```
for (i = 0; i < 4; i++)
```

```
{
```

```
for (j = 0; j < 4; j++)
```

```
{
```

```
cin >> a[i][j];
```

```
}
```

```
}
```



```

for (i=0; i<4; i++)
{
    trace += matrix a[i][j];
}
cout << "Trace of Matrix:" <<
    trace << endl;
    getch();
}

```

other Method.

(VI)

```

#include <iostream.h>
#include <conio.h>
#include <math.h>
void main()
{
    int matrix[3][3], transpose[3][3];
    int i, j;
    cout << "Enter elements of matrix:";
    for (i=0; i<3; i++)
    {
        for (j=0; j<3; j++)

```

```

    }
    cin >> matrix[i][j];
    }
    }
    // calculate transpose
    for (i=0; i<3; i++)
    {
        for (j=0; j<3; j++)
        {
            transpose[j][i] = matrix[i][j];
        }
    }
    cout << "Transpose of matrix:";
    for (i=0; i<3; i++)
    {
        for (j=0; j<3; j++)
        {
            cout << transpose[i][j] << endl;
        }
    }
    getch();
}

```


Long Question:- (I)

```
#include <iostream.h>
```

```
#include <conio.h>
```

```
#include <math.h>
```

```
void main()
```

```
{ float f ( float x, float y)
```

```
{
```

```
float d;
```

```
d =  $x^3 - y$ ;
```

```
return d;
```

```
}
```

```
void main()
```

```
{
```

```
float x0, y0, xup, h, ynew, n;
```

```
cout << "Simple Euler's method" << endl;
```

```
cout << "Enter the value of x0";
```

```
cin >> x0;
```

```
cout << "Enter the value of y0";
```

```
cin >> y0;
```

```
cout << "Enter upper limit of x";
```

```
cin >> xup;
```

cout << "Enter the interval";
cin >> h;

$n = (x_{up} - x_0) / h;$

```
yout = y0 + h * (f(x0, y0) + f(xn, yn)) / 2;  
for (int i = 0; i < n; i++)  
{  
    ynew = y0 + h * f(x0, y0);  
    cout << i << " << x0 << " << y0 << "  
    << " << ynew << endl;  
    y0 = ynew;  
    x0 = x0 + h;  
}  
getch();  
}
```

(iii)

```
#include <iostream.h>  
#include <conio.h>  
#include <math.h>  
void main ()
```



```

{
    clrscr();
    float f(float x, float y);
    float x, yprev, ynext, h, k1, k2,
        k3, k4;
    x0 = 0;
    yprev = 1;
    h = 0.1;
    while(x <= 0.5)
    {
        k1 = f(x, yprev);
        k2 = f(x + h/2, yprev + h * k1/2);
        k3 = f(x + h/2, yprev + h * k2/2);
        k4 = f(x + h, yprev + h * k3);
        ynext = yprev + h * (k1 + 2 * k2 + 2 * k3
            + k4 / 6);
        x = x + h;
        cout << "y (" << x << ") = " <<
            ynext << endl;
        yprev = ynext;
    }
    getch();
}

```

```

float f (float x, float y)
{
    return (x+y);
}

```

(II)

```

#include <iostream.h>
#include <conio.h>
#include <math.h>
void main ()
{
    cout << "Trapezoidal rule":<<endl;
    float h, n, low, up, sum=0;
    cout << "Enter the lower limit"
    << endl;
    cin >> low;
    cout << "Enter the upper limit"
    << endl;
    cin >> up;
    cout << "enter the no. of intervals"
    << endl;
}

```



```
(in) >> n;
```

```
h = (up - low) / n;
```

```
cout << "step size = " << h << endl;
```

```
sum = f(low) + f(up);
```

```
for (int i = 1; i < n; i++)
```

```
{
```

```
sum += 2 * f(low + i * h);
```

```
cout << "I = " << sum << endl;
```

```
getch();
```

```
}
```

```
float f (float x)
```

```
{
```

```
float b;
```

```
b = 1 / x;
```

```
return b;
```

```
}
```
